

CONTACT INFORMATION	Phone: (+91)-9567320446 E-mail: vishnutr1992@gmail.com Website: vishnu-tr.github.io
EMPLOYMENT	Instructor (Physics), <i>Talent Development Center</i>, Indian Institute of Science, Challakere Campus (May 2026 - Present). Govt. Guest Lecturer, Department of Physics, <i>MAMO College</i>, Manassery (July 2025 - March 2026). Postdoctoral Fellow, <i>Raman Research Institute</i>, Mentor: <i>Prof. Dibyendu Roy</i> (May 2022 -May 2025). Research Associate, <i>Chennai Mathematical Institute</i>, Mentor: <i>Prof. Govind S Krishnaswami</i> (October 2021 - March 2022).
EDUCATION	Ph.D <i>August 2016-September 2021</i> <i>Chennai Mathematical Institute</i>, Ph.D. in Physics, Thesis title: Integrability and dynamics of the Rajeev-Ranken model, Advisor: <i>Prof. Govind S Krishnaswami</i>. Masters <i>August 2014-July 2016</i> <i>Chennai Mathematical Institute</i>, M.Sc. in Physics, Thesis Title: Integrability and inverse scattering transform for the KdV equation, Advisor: <i>Prof. Govind S Krishnaswami</i>. Masters <i>July 2012-May 2014</i> <i>University of Hyderabad</i>, M.Sc. in Physics, Thesis Title: Study of Neutrino mass models, Advisor: <i>Prof. Rukmani Mohanta</i>. Undergraduate <i>August 2009-July 2012</i> <i>St. Joseph's College, Devagiri</i>, B.Sc. Physics, Project Title: Spectroscopic studies of the twin quasar 0957+561 - The first gravitational lens (Group project), Advisor: <i>Prof. S I Issac</i>.
PUBLICATIONS	• <i>Nonlinearity-induced transition in heat conduction through a topological metamaterial of rotors</i>, T R Vishnu and Dibyendu Roy, arXiv:2511.14560 [cond-mat.stat-mech]. • <i>From Chiral Topological Dynamics to Chiral Topological Amplification: Real vs Imaginary Parameters in a Hermitian Bosonic Chain</i>, Kiran B Estake, T R Vishnu and Dibyendu Roy, <i>Phys. Rev. B</i> 112, 214314 (2025), arXiv:2508.14560 [quant-ph]. • <i>Heat transport through an open coupled scalar field theory hosting stability-to-instability transition</i>, T R Vishnu and Dibyendu Roy, <i>J. Stat. Phys.</i> 191, 123 (2024), arXiv:2402.04986 [cond-mat.stat-mech]. • <i>Screwon spectral statistics and dispersion relation in the quantum Rajeev-Ranken model</i>, Govind S Krishnaswami and T R Vishnu, <i>Physica D: Nonlinear Phenomena</i> 463, 134170 (2024), arXiv:2312.13122 [nlin.SI]. • <i>Spectral solutions for the Schrödinger equation with a regular singularity</i>, Pushkar Mohile, Ayaz Ahmed, T R Vishnu, and Pichai Ramadevi, <i>SciPost Phys. Core</i> 7, 041 (2024), arXiv:2309.00026 [quant-ph]. • <i>Quantum Rajeev-Ranken model as an anharmonic oscillator</i>, Govind S Krishnaswami and T R Vishnu, <i>J. Math. Phys.</i> 63, 032101 (2022), arXiv:2111.03858 [math-ph].

- *The idea of a Lax pair-Part II: Continuum wave equations*, Govind S Krishnaswami and T R Vishnu, [Resonance](#) **26**, 257 (2021).
- *The idea of a Lax pair-Part I: Conserved quantities for a dynamical system*, Govind S Krishnaswami and T R Vishnu, [Resonance](#) **25**, 1705 (2020).
- *An introduction to Lax pairs and the zero curvature representation*, Govind S Krishnaswami and T R Vishnu, [arXiv:2004.05791\[nlin.SI\]](#).
- *Invariant tori, action-angle variables and phase space structure of the Rajeev-Ranken model*, Govind S Krishnaswami and T R Vishnu, [J. Math. Phys.](#) **60**, 082902 (2019), [arXiv:1906.03141\[nlin.SI\]](#).
- *On the Hamiltonian formulation, integrability and algebraic structures of the Rajeev-Ranken model*, Govind S Krishnaswami and T R Vishnu, [J. Phys. Commun.](#) **3**, 025005 (2019), [arXiv:1804.02859 \[hep-th\]](#).

RESEARCH INTERESTS

Theoretical and Mathematical Physics: Quantum field theory, Integrability in classical and quantum systems and field theories, Dynamical systems, Heat transport, Open quantum systems, Random matrix theory, Nonlinear dynamics & Chaos.

RESEARCH EXPERIENCE

- Two dimensional field theory, Partial differential equations, Poisson-Lie algebras, Inverse scattering, KdV equation, Lax pairs, r -matrices, Hamiltonian formulation, Integrability, Invariant tori and action-angle variables, Exact-WKB method, Energy-level statistics, Quantum Langevin equations, Non-equilibrium Green's functions, OTOC.

CONFERENCES AND SCHOOLS

- *9th Indian Statistical Physics Community Meeting*, 3-5 April, 2024, International Center for Theoretical Sciences, Bangalore.
- *Stability of quantum matter in and out of equilibrium at various scales*, 15-26 January, 2024, International Center for Theoretical Sciences, Bangalore.
- *8th Indian Statistical Physics Community Meeting*, 1-3 February, 2023, International Center for Theoretical Sciences, Bangalore.
- *Conference on Nonlinear Systems and Dynamics* (Online), 17-22 December, 2021, SASTRA Deemed University, Thanjavur.
- *Bangalore School on Statistical Physics - XII* (Online), 28 June-9 July, 2021, International Center for Theoretical Sciences, Bangalore.
- *Lecture series on Basics of nonlinear integrable systems and their applications* (Online), 7-17 April, 2021, SASTRA Deemed University, Thanjavur.
- *XXXIII SERB Main school-Theoretical High Energy Physics*, 7-26 December, 2019, S.G.T.B. Khalsa College, University of Delhi.
- *Young Researchers Integrability School and Workshop: A modern primer for 2D CFT*, 10-16 February, 2019, Erwin Schrödinger international Institute of Mathematics and Physics, Vienna.
- *Conference on Nonlinear Systems and Dynamics*, 11-14 October, 2018, Jawaharlal Nehru University - New Delhi.
- *Integrable systems in Mathematics, Condensed Matter and Statistical Physics*, 16 July-10 August, 2018, International Center for Theoretical Sciences, Bangalore.

TALKS AND POSTERS

- Talk, *Heat transport: Topological metamaterial of rotors*, IN-House 2025, May 15, 2025, Raman Research Institute, Bangalore.
- Talk, *Heat transport through an open coupled scalar field theory hosting stability-to-instability transition*, IN-House 2024, May 8, 2024, Raman Research Institute, Bangalore.
- Talk, *Heat transport through an open coupled scalar field theory hosting stability-to-instability transition*, 9th Indian Statistical Physics Community Meeting, April 05, 2024, International Center for Theoretical Sciences, Bangalore.
- Poster Presentation, *Signature of stability-to-instability transitions in heat trans-*

port through a coupled scalar field theory, Stability of quantum matter in and out of equilibrium at various scales, January 24, 2024, International Center for Theoretical Sciences, Bangalore.

- Talk, *Probing dynamical stability-to-instability transition in a network by heat conduction*, IN-House 2023, April 14, 2023, Raman Research Institute, Bangalore.
- Talk, *Dynamical stability of a coupled scalar field theory: Different perspectives*, 8th Indian Statistical Physics Community Meeting, February 03, 2023, International Center for Theoretical Sciences, Bangalore.
- Talk, *Dynamics and integrability of the Rajeev-Ranken model*, Chennai Strings Meeting, December 15, 2020, Institute of Mathematical Sciences (Webinar).
- Talk, *Integrability of a mechanical reduction of a dual to the principal chiral model*, National Symposium on Theoretical High Energy Physics, December 20, 2019, SGTB Khalsa College, University of Delhi.
- CMI Seminar, *On the Hamiltonian formulation and integrability of the Rajeev-Ranken model*, Chennai Mathematical Institute, March 20, 2019, Chennai.
- Poster presentation, *Hamiltonian dynamics and integrability of the Rajeev-Ranken model*, Conference on Nonlinear Systems and Dynamics, 11-14 October, 2018, Jawaharlal Nehru University, New Delhi.
- Poster presentation, *Hamiltonian dynamics and integrability of the Rajeev-Ranken model*, Integrable systems in Mathematics, Condensed matter and Statistical Physics, 16 July-10 August, 2018, International Center for Theoretical Sciences, Bangalore.
- CMI Seminar, *Some features of a 1+1 dimensional field theory dual to the Principal Chiral model*, Chennai Mathematical Institute, October 3, 2017, Chennai.

TEACHING

- Teaching assistant for the course on Classical Mechanics, Course Instructor: Govind S Krishnaswami, Sep-Dec 2021, Chennai Mathematical Institute.
- Teaching assistant for the course on Thermal Physics, Course Instructor: Govind S Krishnaswami, Aug-Nov 2019, Chennai Mathematical Institute.
- Teaching assistant for the Workshop of the Academy of Physics Teachers, Kerala, Topic: Scattering in Quantum Mechanics, Course Instructor: Govind S Krishnaswami, 23-24 June, 2018, Christ College, Irinjalakuda.
- Teaching assistant for the course on Continuum Mechanics, Course Instructor: Govind S Krishnaswami, Jan-Apr 2018, Chennai Mathematical Institute.

INTERNSHIPS

Institute of Physics

May-June 2013

Students' summer visiting program (SSVP-2013)- A reading project on 'Neutrino oscillations' under the guidance of Prof. Pankaj Agarwal at IOP Bhubaneswar.

ACHIEVEMENTS

- 2011 - National Cadet Corps - C certificate holder
- 2012 - Was ranked among the top 15 in all Kerala Physics Talent Search, conducted by Academy of Physics Teachers, (APT) Kerala.
- 2014 - Qualified Joint Entrance Screening Test (JEST).

REFERENCES

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